

Unit 1

Digitalization of the Global Economy

Framework Library · 20 reference sheets

1.1**Digital Transformation**

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LEGEND

DEFINITION	Conceptual definitions and key terms (8 cards)
FRAMEWORK	Strategic frameworks and analytical tools (12 cards)
K	

Atoms → Bits

SLIBRARY 01 · SHEET
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Negroponete's foundational shift — from physical substrate to informational substrate.

PROFESSIONAL DEFINITION

Nicholas Negroponete's framing of the central economic transition of the late 20th and early 21st centuries: the displacement of atoms (physical goods, physical distribution, physical interaction) by bits (digital goods, digital distribution, digital interaction) as the dominant substrate of value creation.

WHO DEVELOPED IT

Negroponete, N. (1995) 'Being Digital', New York: Knopf.

WHY IT WAS DEVELOPED

Foundational concept of digital-economy thinking; preceded and shaped by Toffler's 'The Third Wave' (1980) and Castells' 'The Information Age' (1996).

FIGURE 1 · ATOMS → BITS

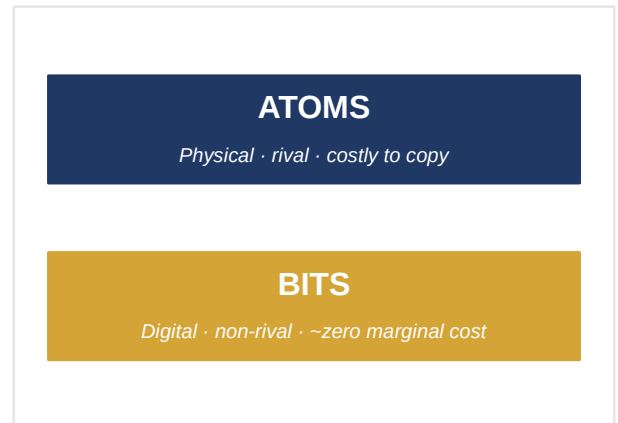


Figure 1. Atoms → Bits: Negroponete's foundational shift — from physical substrate to informational substrate.

HOW IT IS USED

when explaining the foundational economic logic of any digital business.

WHEN IT IS USED

when scoping an industry digitisation analysis.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Transformation.

REAL-WORLD EXAMPLES

● Music: vinyl/CD → MP3 → streaming (Spotify) — full atoms-to-bits transition.

● Photography: film → digital files → cloud → AI-generated.

● Banking: branch transaction → online → mobile-app-only neobanks.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Atoms → Bits contributes a clearer, more transferable framing of the dynamic it names — Negroponete's foundational shift — from physical substrate to informational substrate. Its analytical contribution is the explicit recognition that bits replace atoms wherever the underlying value is informational rather than physical.

RELATED FRAMEWORKS IN THIS COURSE

Digitisation · Digitalisation · DT · 1.1.5

Zero Marginal Cost (Unit 2.4.1)

Christensen's Disruption · 1.4.1

SOURCES (HARVARD REFERENCING STYLE)

Negroponete, N. (1995) Being Digital. New York: Knopf.

Toffler, A. (1980) The Third Wave. New York: Bantam Books.

Castells, M. (1996) The Information Age: Economy, Society, and Culture. Oxford: Blackwell.

OECD 3-Layer Model

Three nested scopes for measuring the digital economy.

PROFESSIONAL DEFINITION

An OECD-developed measurement framework for the digital economy that recognises three nested layers: (1) the core ICT sector itself; (2) digitalising sectors (firms whose primary product isn't digital but whose operations are heavily digital); (3) the digital economy at large, where digital is integral to the value proposition.

WHO DEVELOPED IT

OECD (2020) 'A Roadmap toward a Common Framework for Measuring the Digital Economy', G20 Saudi Arabia.

WHY IT WAS DEVELOPED

Earlier conceptualisation in Bukht, R. and Heeks, R. (2017) 'Defining, Conceptualising and Measuring the Digital Economy', University of Manchester.

FIGURE 1 · OECD 3-LAYER MODEL

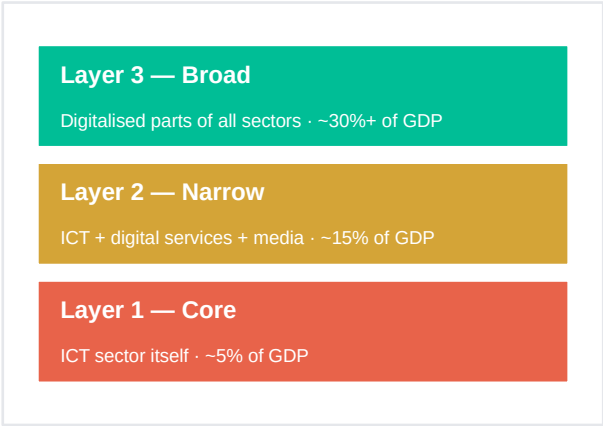


Figure 1. OECD 3-Layer Model: Three nested scopes for measuring the digital economy.

HOW IT IS USED

when scoping any quantitative claim about 'the digital economy'.

WHEN IT IS USED

in policy and benchmarking work where measurement comparability matters.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Transformation.

REAL-WORLD EXAMPLES

- OECD Digital Economy Outlook annual reports — explicit use of all three layers.
- UNCTAD Digital Economy Report 2024 — uses Bukht-Heeks-derived scope.
- EU Digital Economy and Society Index (DESI) — measures Layer 2–3 components.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, OECD 3-Layer Model contributes a clearer, more transferable framing of the dynamic it names — Three nested scopes for measuring the digital economy. Its analytical contribution is the explicit recognition that layer 1 — Core: information and communication technology (ICT) sector — telecom, software, hardware, IT services.

RELATED FRAMEWORKS IN THIS COURSE

- Atoms → Bits · 1.1.1
- UNCTAD Digital Trade Flows · 1.2.1
- ICT Sector % of GDP · 1.2.2

SOURCES (HARVARD REFERENCING STYLE)

OECD (2020) A Roadmap Toward a Common Framework for Measuring the Digital Economy. Paris: OECD Publishing.

Bukht, R. and Heeks, R. (2017) 'Defining, conceptualising and measuring the digital economy', Centre for Development Informatics Working Paper 68. Manchester: University of Manchester.

OECD (2024) OECD Digital Economy Outlook 2024. Paris: OECD Publishing.

S-Curve of Adoption

SLIBRARY 01 · SHEET
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Slow start · exponential rise · saturation plateau — the universal diffusion shape.

PROFESSIONAL DEFINITION

The S-curve is the canonical shape of technology diffusion through a population: a slow innovator-and-early-adopter phase, followed by a steep early-majority and late-majority climb, ending in a saturation plateau as laggards eventually adopt or the market is exhausted. The shape recurs across electricity, telephones, internet, smartphones, and AI.

WHO DEVELOPED IT

Rogers, E.M. (1962, 5th edn 2003) 'Diffusion of Innovations', Free Press.

WHY IT WAS DEVELOPED

Built on Tarde's 'imitation' theory (1903) and Bass's diffusion model (1969).

FIGURE 1 · S-CURVE OF ADOPTION

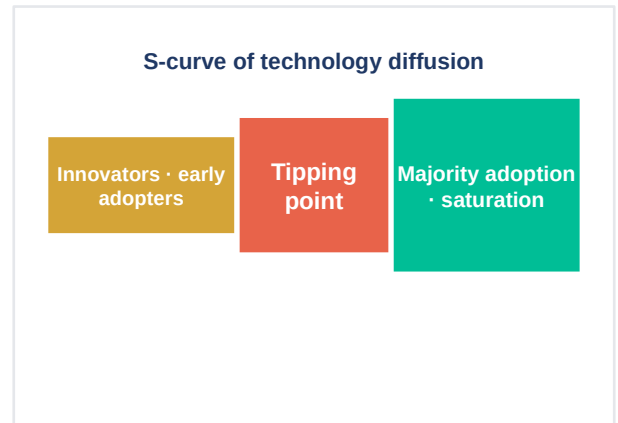


Figure 1. S-Curve of Adoption: Slow start · exponential rise · saturation plateau — the universal diffusion shape.

HOW IT IS USED

when assessing where a technology sits in its diffusion arc.

WHEN IT IS USED

in marketing and product strategy (whom are we selling to next?).

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Transformation.

REAL-WORLD EXAMPLES

● Smartphones: ~2007 launch → ~2015 majority → ~2024 saturation in OECD countries.

● Generative AI 2022–2025: rapid early-adopter and early-majority climb.

● Electric vehicles: ~2025 transitioning from early adopters to early majority in Europe.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, S-Curve of Adoption contributes a clearer, more transferable framing of the dynamic it names — Slow start · exponential rise · saturation plateau — the universal diffusion shape. Its analytical contribution is the explicit recognition that phase 1 — Innovators (~2.5%): risk-tolerant, technically curious, often researchers or hobbyists.

RELATED FRAMEWORKS IN THIS COURSE

[Gartner Hype Cycle · 1.1.4](#)[Christensen's Disruption · 1.4.1](#)[Industry Digitization Index · 1.4.2](#)

SOURCES (HARVARD REFERENCING STYLE)

Rogers, E.M. (2003) Diffusion of Innovations. 5th edn. New York: Free Press.

Bass, F.M. (1969) 'A new product growth for model consumer durables', Management Science, 15(5), pp. 215–227.

Geroski, P.A. (2000) 'Models of technology diffusion', Research Policy, 29(4–5), pp. 603–625.

Gartner Hype Cycle

SLIBRARY 01 · SHEET
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Trigger · peak · trough · slope · plateau — the emotional shape of technology adoption.

PROFESSIONAL DEFINITION

Gartner's five-phase model of expectation dynamics around an emerging technology: Innovation Trigger → Peak of Inflated Expectations → Trough of Disillusionment → Slope of Enlightenment → Plateau of Productivity. Distinct from the S-curve in that it tracks public/expectation attitude rather than adoption percentage.

WHO DEVELOPED IT

Gartner Inc. (1995–) annual Hype Cycle reports.

WHY IT WAS DEVELOPED

Conceptual lineage: behavioural-finance studies of asset-price overshoot (Shiller, 2000) and innovation-management literature (Roehrich, 2009).

FIGURE 1 · GARTNER HYPE CYCLE

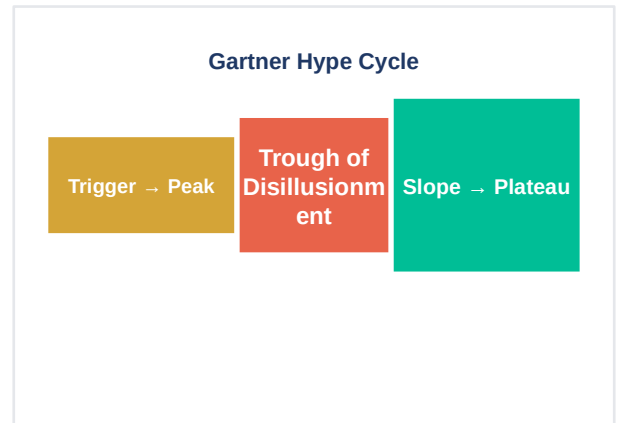


Figure 1. Gartner Hype Cycle: Trigger · peak · trough · slope · plateau — the emotional shape of technology adoption.

HOW IT IS USED

when assessing a buzzword-heavy emerging technology.

WHEN IT IS USED

in venture and corporate strategy investment decisions.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Transformation.

REAL-WORLD EXAMPLES

● Crypto / NFTs: trigger 2009 → peak 2021 → trough 2022–2024.

● Generative AI: trigger 2017 (Transformers) → peak 2023 → entering trough/slope 2025.

● Metaverse: peak 2021 → deep trough 2023+.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Gartner Hype Cycle contributes a clearer, more transferable framing of the dynamic it names — Trigger · peak · trough · slope · plateau — the emotional shape of technology adoption. Its analytical contribution is the explicit recognition that phase 1 — Innovation trigger: a breakthrough creates initial press interest; few real products exist.

RELATED FRAMEWORKS IN THIS COURSE

S-Curve of Adoption · 1.1.3

Christensen's Disruption · 1.4.1

Step-Function AI Impact (Unit

SOURCES (HARVARD REFERENCING STYLE)

Fenn, J. and Raskino, M. (2008) Mastering the Hype Cycle: How to Choose the Right Innovation at the Right Time. Boston: Harvard Business Press.

Gartner Inc. (2024) Gartner Hype Cycle for Emerging Technologies 2024. Stamford, CT: Gartner.

Roehrich, G. (2009) 'Hype cycle as a marketing tool', Journal of Marketing Theory and Practice, 17(3), pp. 235–246.

Digitisation · Digitalisation · DT

SLIBRARY 01 · SHEET
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Three distinct concepts often confused — artefact change · process change · business-model change.

PROFESSIONAL DEFINITION

Three frequently conflated concepts: digitisation (converting an analogue artefact to digital form — e.g. scanning a paper file), digitalisation (using digital technology to change a process — e.g. moving claims-handling online), and digital transformation (changing the business model itself — e.g. a record label becoming a streaming-and-data company).

WHO DEVELOPED IT

Distinction first articulated in Brennen, J.S. and Kreiss, D. (2014) 'Digitalization and Digitization', International Encyclopedia of Communication.

WHY IT WAS DEVELOPED

Refined in Verhoef, P.C. et al. (2021) 'Digital transformation: A multidisciplinary reflection and research agenda', Journal of Business Research.

FIGURE 1 · DIGITISATION · DIGITALISATION · DT

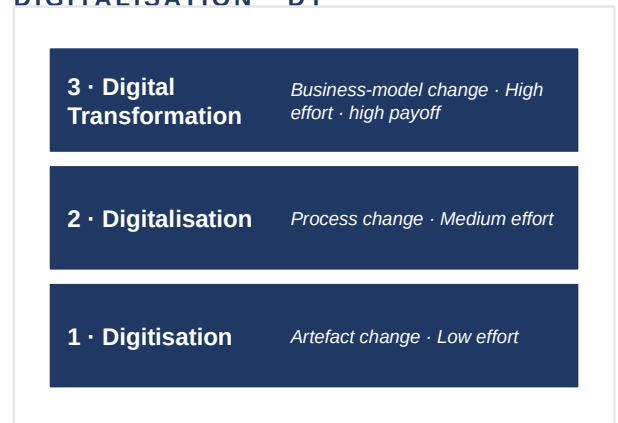


Figure 1. Digitisation · Digitalisation · DT: Three distinct concepts often confused — artefact change · process change · business-model change.

HOW IT IS USED

in any conversation about 'going digital' — clarifies which kind of change is meant.

WHEN IT IS USED

to challenge unrealistic transformation claims.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Digital Transformation.

REAL-WORLD EXAMPLES

● Digitisation: scanning paper invoices into PDFs.

● Digitalisation: replacing paper invoices with an online billing system.

● Transformation: a software firm pivoting from licence sales to SaaS subscription with usage-based pricing.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Digitisation · Digitalisation · DT contributes a clearer, more transferable framing of the dynamic it names — Three distinct concepts often confused — artefact change · process change · business-model change. Its analytical contribution is the explicit recognition that digitisation: artefact-level change. Output = digital artefact. Effort = low.

RELATED FRAMEWORKS IN THIS COURSE

Atoms → Bits · 1.1.1

BCG 10-20-70 Rule · 1.4.3

Christensen's Disruption · 1.4.1

SOURCES (HARVARD REFERENCING STYLE)

Brennen, J.S. and Kreiss, D. (2014) 'Digitalization and digitization', in The International Encyclopedia of Communication. Oxford: Wiley-Blackwell.

Verhoef, P.C., Broekhuizen, T., Bart, Y., Bhattacharya, A., Dong, J.Q., Fabian, N. and Haenlein, M. (2021) 'Digital transformation: a multidisciplinary reflection and research agenda', Journal of Business Research, 122, pp. 889–901.

Gartner Inc. (2024) IT Glossary: Digital Transformation. Stamford, CT: Gartner.

UNCTAD Digital Trade Flows

Cross-border data flows now exceed traditional trade in many sectors.

PROFESSIONAL DEFINITION

UNCTAD's framework for measuring international digital trade — services delivered electronically, digital intermediation of physical trade, and pure cross-border data flows. The headline finding: digital and digitally-deliverable services have grown roughly twice as fast as goods trade since 2010, and now account for over half of global services trade.

WHO DEVELOPED IT

UNCTAD (2024) Digital Economy Report 2024: Shaping an Environmentally Sustainable and Inclusive Digital Future.

WHY IT WAS DEVELOPED

Earlier methodology in UNCTAD (2019, 2021) Digital Economy Report.

FIGURE 1 · UNCTAD DIGITAL TRADE FLOWS



Figure 1. UNCTAD Digital Trade Flows: Cross-border data flows now exceed traditional trade in many sectors.

HOW IT IS USED

when constructing the size and growth case for the digital economy.

WHEN IT IS USED

in trade-policy and tax-policy analysis.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Economy Data.

REAL-WORLD EXAMPLES

- Cloud services: AWS, Azure, GCP cross-border revenues classified under digitally-deliverable services.
- Streaming: Netflix, Spotify cross-border subscriber revenues.
- Data localisation regulations (China, Russia, India) directly target the data-flow component.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, UNCTAD Digital Trade Flows contributes a clearer, more transferable framing of the dynamic it names — Cross-border data flows now exceed traditional trade in many sectors. Its analytical contribution is the explicit recognition that three measurement categories: digitally-deliverable services · digital-platform-intermediated trade · pure data flows.

RELATED FRAMEWORKS IN THIS COURSE

- OECD 3-Layer Model · 1.1.2
- ICT Sector % of GDP · 1.2.2
- DCO Regional Digital Trends · 1.2.4

SOURCES (HARVARD REFERENCING STYLE)

UNCTAD (2024) Digital Economy Report 2024. Geneva: United Nations Conference on Trade and Development.

OECD (2024) OECD Digital Economy Outlook 2024. Paris: OECD Publishing.

WTO (2024) World Trade Statistical Review 2024. Geneva: World Trade Organization.

ICT Sector % of GDP

OECD methodology — measuring the digital sector's contribution to national output.

PROFESSIONAL DEFINITION

The OECD methodology for measuring the ICT (Information and Communication Technology) sector's contribution to national GDP. The ICT sector is defined by ISIC Rev. 4 codes covering hardware manufacturing, software publishing, telecommunications, and IT services. Direct contribution to OECD GDP: ~5–7% on average; in leading economies (USA, Korea, Ireland) 8–14%.

WHO DEVELOPED IT

OECD (2011, updated 2020) 'Guide to Measuring the Information Society'.

WHY IT WAS DEVELOPED

Built on the 1998 OECD/Statistics Canada definition of the ICT sector. ISIC Rev. 4 codes 26, 58, 61, 62, 63 form the operational definition.

FIGURE 1 · ICT SECTOR % OF GDP



Figure 1. ICT Sector % of GDP: OECD methodology — measuring the digital sector's contribution to national output.

HOW IT IS USED

as a baseline benchmark for cross-country digital-economy comparison.

WHEN IT IS USED

in industrial-policy and FDI-attraction analysis.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Economy Data.

REAL-WORLD EXAMPLES

- Ireland: high share due to MNC tax-driven location of digital firms (Google, Meta, Apple Europe).
- Korea: organic high share — Samsung, LG, SK Hynix anchor the ICT sector.
- Estonia: small absolute economy but high digital share via e-Residency and digital government.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, ICT Sector % of GDP contributes a clearer, more transferable framing of the dynamic it names — OECD methodology — measuring the digital sector's contribution to national output. Its analytical contribution is the explicit recognition that includes: hardware manufacturing (semiconductors, computers, networking), software publishing, telecoms, computer services, data services.

RELATED FRAMEWORKS IN THIS COURSE

- OECD 3-Layer Model · 1.1.2
- UNCTAD Digital Trade Flows · 1.2.1
- GAFAM Concentration (Unit 3.1)

SOURCES (HARVARD REFERENCING STYLE)

OECD (2011) Guide to Measuring the Information Society 2011. Paris: OECD Publishing.

OECD (2024) OECD Digital Economy Outlook 2024. Paris: OECD Publishing.

United Nations (2008) International Standard Industrial Classification of All Economic Activities, Rev. 4. New York: UN Statistics Division.

Global Internet Penetration

ITU 2024 — 5.5bn online · gaps remain across regions and demographics.

PROFESSIONAL DEFINITION

The percentage of the global population using the internet, tracked annually by the International Telecommunication Union (ITU). 2024 estimate: ~67% of the global population, or ~5.5 billion users, with substantial regional variation: Europe ~92%, Americas ~88%, Asia-Pacific ~71%, Africa ~37%. Gender, urban-rural, and income gaps persist.

WHO DEVELOPED IT

ITU (annual since 1996) Measuring Digital Development: Facts and Figures.

WHY IT WAS DEVELOPED

Methodology defined in ITU Handbook for Measuring ICT Access and Use by Households and Individuals (2020).

FIGURE 1 · GLOBAL INTERNET PENETRATION

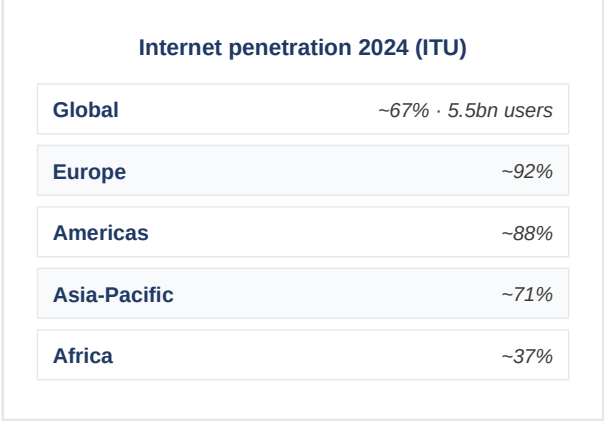


Figure 1. Global Internet Penetration: ITU 2024 — 5.5bn online · gaps remain across regions and demographics.

HOW IT IS USED

when scoping the addressable market for any digital business, especially in emerging markets.

WHEN IT IS USED

in development-policy and digital-divide analysis.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Economy Data.

REAL-WORLD EXAMPLES

- Reliance Jio (India) launch 2016: drove India's penetration from ~30% to ~50% in two years.
- Starlink: targeting unconnected rural users at scale 2022+.
- EU Digital Decade target: 100% broadband-coverage gigabit by 2030.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Global Internet Penetration contributes a clearer, more transferable framing of the dynamic it names — ITU 2024 — 5.5bn online · gaps remain across regions and demographics. Its analytical contribution is the explicit recognition that headline metric: % of population using the internet in the past three months.

RELATED FRAMEWORKS IN THIS COURSE

- The Digital Divide Gap · 1.2.5
- DCO Regional Digital Trends · 1.2.4
- UNCTAD Digital Trade Flows · 1.2.1

SOURCES (HARVARD REFERENCING STYLE)

ITU (2024) Measuring Digital Development: Facts and Figures 2024. Geneva: International Telecommunication Union.

ITU (2020) Handbook for Measuring ICT Access and Use by Households and Individuals. Geneva: ITU.

World Bank (2024) Digital Progress and Trends Report 2024. Washington, DC: World Bank.

DCO Regional Digital Trends

Digital Cooperation Organization — adoption patterns across MENA, Asia, and Africa.

PROFESSIONAL DEFINITION

The Digital Cooperation Organization (DCO) is an intergovernmental body of 16 member states (largely MENA, parts of Asia and Africa) that publishes annual Digital Economy Trends reports tracking digital-economy progress across emerging markets. Useful complement to OECD/UNCTAD coverage, which is biased toward developed economies.

WHO DEVELOPED IT

DCO founded 2020 (Riyadh, Saudi Arabia).

WHY IT WAS DEVELOPED

Member states include Bahrain, Cyprus, Djibouti, Greece, Jordan, Kuwait, Morocco, Nigeria, Oman, Pakistan, Qatar, Rwanda, Saudi Arabia, Senegal, Sri Lanka, The Gambia. Annual Digital Economy Trends reports since 2022.

FIGURE 1 · DCO REGIONAL DIGITAL TRENDS

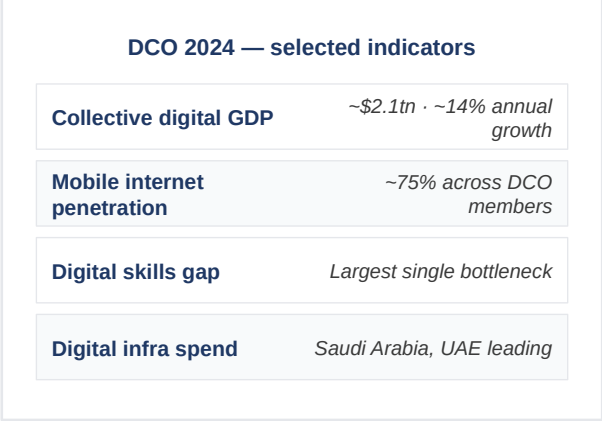


Figure 1. DCO Regional Digital Trends: Digital Cooperation Organization — adoption patterns across MENA, Asia, and Africa.

HOW IT IS USED

when analysing digital-economy strategy in MENA / non-OECD geographies.

WHEN IT IS USED

as cross-check against OECD/UNCTAD figures.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Digital Economy Data.

REAL-WORLD EXAMPLES

- Saudi Vision 2030: explicit digital-economy targets aligned with DCO framework.
- M-Pesa (Kenya): mobile-money leapfrogging documented in DCO regional reports.
- UAE Digital Economy Strategy 2030: 20% of non-oil GDP target.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, DCO Regional Digital Trends contributes a clearer, more transferable framing of the dynamic it names — Digital Cooperation Organization — adoption patterns across MENA, Asia, and Africa. Its analytical contribution is the explicit recognition that tracks: digital infrastructure, digital skills, digital business environment, digital governance, regional cooperation.

RELATED FRAMEWORKS IN THIS COURSE

- OECD 3-Layer Model · 1.1.2
- Global Internet Penetration · 1.2.3
- The Digital Divide Gap · 1.2.5

SOURCES (HARVARD REFERENCING STYLE)

- Digital Cooperation Organization (2024) DCO Digital Economy Trends 2024. Riyadh: DCO.
- Digital Cooperation Organization (2025) DCO Digital Economy Trends 2025. Riyadh: DCO.
- World Bank (2024) Digital Progress and Trends Report 2024. Washington, DC: World Bank.

The Digital Divide Gap

The persistent inequality of access · skills · and use that shadows digital growth.

PROFESSIONAL DEFINITION

The digital divide is the gap between those who have effective access to and capability with digital technology and those who do not. It operates along three dimensions: access (do you have the device and connection?), skills (can you use it productively?), and use (do you actually use it for high-value activities?). The first-order divide closes; the second- and third-order persist.

WHO DEVELOPED IT

Term in academic use since DiMaggio, P. and Hargittai, E. (2001) 'From the digital divide to digital inequality', Princeton CCSRP Working Paper.

WHY IT WAS DEVELOPED

Three-level framing in van Dijk, J. (2005) 'The Deepening Divide: Inequality in the Information Society'.

FIGURE 1 · THE DIGITAL DIVIDE GAP

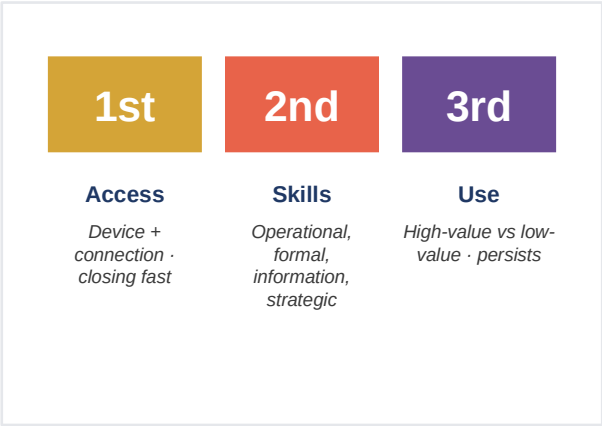


Figure 1. The Digital Divide Gap: The persistent inequality of access · skills · and use that shadows digital growth.

HOW IT IS USED

in any policy or strategy that targets 'underserved' digital users.

WHEN IT IS USED

in CSR and inclusion programme design.

BY WHOM IT IS USED

educators, analysts, and practitioners requiring a precise definitional anchor when scoping their work in Digital Economy Data.

REAL-WORLD EXAMPLES

- EU Digital Decade goal: 80% of adults with basic digital skills by 2030 (currently ~55%).
- Generative AI use by socioeconomic status (Pew 2024): ~3x gap between top and bottom income deciles.
- South Korea: highest internet penetration globally yet measurable third-order divide by age.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, The Digital Divide Gap contributes a clearer, more transferable framing of the dynamic it names — The persistent inequality of access · skills · and use that shadows digital growth. Its analytical contribution is the explicit recognition that first-order divide (access): physical access to a device and a connection. Closing fast — but not closed.

RELATED FRAMEWORKS IN THIS COURSE

- Global Internet Penetration · 1.2.3
- DCO Regional Digital Trends · 1.2.4
- Skills Matrix 2025-2030 (Unit

SOURCES (HARVARD REFERENCING STYLE)

DiMaggio, P. and Hargittai, E. (2001) 'From the digital divide to digital inequality: studying internet use as penetration increases', Princeton CCSRP Working Paper, 15.

van Dijk, J. (2005) The Deepening Divide: Inequality in the Information Society. Thousand Oaks, CA: Sage.

ITU (2024) Measuring Digital Development: Facts and Figures 2024. Geneva: ITU.

Platform vs. Pipeline

SLIBRARY 03 · SHEET
1 / 5

Two architectures · two strategic playbooks.

PROFESSIONAL DEFINITION

Parker, Van Alstyne and Choudary's distinction between two business architectures: the pipeline (linear value chain — produce, then distribute, then consume) and the platform (orchestrating value-creating interactions between producers and consumers it does not employ). The shift from pipeline to platform is the dominant structural pattern of the digital economy.

WHO DEVELOPED IT

Parker, G., Van Alstyne, M. and Choudary, S.P. (2016) 'Platform Revolution', W.W.

WHY IT WAS DEVELOPED

Norton. Conceptual roots: Rochet & Tirole (2003) two-sided markets; Iansiti & Levien (2004) ecosystem economics.

FIGURE 1 · PLATFORM VS. PIPELINE

Pipeline vs. Platform		
Value chain	Linear · firm-controlled	Networked · ecosystem-mediated
Production	Firm produces	Producers external
Moat	Scale + distribution	Network effects + data
Examples	Toyota · Maersk	Amazon · Uber · YouTube

Figure 1. Platform vs. Pipeline: Two architectures · two strategic playbooks.

HOW IT IS USED

when classifying any digital business model.

WHEN IT IS USED

in strategy work assessing whether to platformise an existing pipeline.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Technology & Ecosystems.

REAL-WORLD EXAMPLES

● Amazon: pipeline (1P retail) and platform (3P Marketplace) coexist.

● Apple: hybrid — pipeline (hardware) + platform (App Store).

● Uber: pure platform — owns no cars, employs no drivers.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Platform vs. Pipeline contributes a clearer, more transferable framing of the dynamic it names — Two architectures · two strategic playbooks. Its analytical contribution is the explicit recognition that pipelines own the value-creation chain — design, build, market, distribute.

RELATED FRAMEWORKS IN THIS COURSE

[Network Effects · 1.3.4](#)
[Multi-Sided Platforms \(Unit 3\)](#)
[BCG 10-20-70 Rule · 1.4.3](#)

SOURCES (HARVARD REFERENCING STYLE)

Parker, G.G., Van Alstyne, M.W. and Choudary, S.P. (2016) Platform Revolution. New York: W.W. Norton.

Iansiti, M. and Levien, R. (2004) The Keystone Advantage. Boston: HBR Press.

Cusumano, M.A., Gawer, A. and Yoffie, D.B. (2019) The Business of Platforms. New York: HarperBusiness.

SMAC Stack

SLIBRARY 03 · SHEET
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Social · Mobile · Analytics · Cloud — the substrate of digital business.

PROFESSIONAL DEFINITION

An umbrella label for four mutually-reinforcing technology layers that, in combination, form the operational substrate of post-2010 digital business: Social (online networks and identity), Mobile (smartphones, apps, ubiquitous connectivity), Analytics (data-driven decisioning), and Cloud (elastic, on-demand compute and storage).

WHO DEVELOPED IT

Term in industry use ~2012 (Cognizant, Gartner, McKinsey research).

WHY IT WAS DEVELOPED

Conceptual integration in Cognizant (2013) 'The Future of Work: Powered by SMAC'. AI is now often added as a fifth layer (SMACI / SMAC+AI).

FIGURE 1 · SMAC STACK

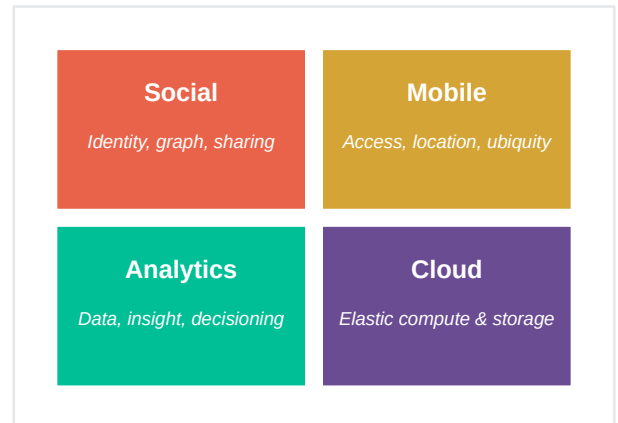


Figure 1. SMAC Stack: Social · Mobile · Analytics · Cloud — the substrate of digital business.

HOW IT IS USED

when scoping a digital architecture or transformation programme.

WHEN IT IS USED

in IT-strategy and capability assessment.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Technology & Ecosystems.

REAL-WORLD EXAMPLES

Modern banking app: cloud-hosted, mobile-delivered, social-shared (e.g. Revolut bill-splitting), analytics-powered fraud detection.

Uber: SMAC-native business — couldn't have existed pre-2009.

Healthcare digitisation 2025: SMAC + AI converging into 'SMACI'.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, SMAC Stack contributes a clearer, more transferable framing of the dynamic it names — Social · Mobile · Analytics · Cloud — the substrate of digital business. Its analytical contribution is the explicit recognition that cloud — the foundation: elastic compute, storage, and networking on demand.

RELATED FRAMEWORKS IN THIS COURSE

Platform vs. Pipeline · 1.3.1

Cloud: IaaS · PaaS · SaaS (Uni

AI Spectrum (Unit 4.3.1).

SOURCES (HARVARD REFERENCING STYLE)

Cognizant Center for the Future of Work (2013) The Future of Work: Powered by SMAC. Teaneck, NJ: Cognizant.

Gartner Inc. (2013) The Nexus of Forces: Social, Mobile, Cloud and Information. Stamford, CT: Gartner.

Brynjolfsson, E. and McAfee, A. (2014) The Second Machine Age. New York: W.W. Norton.

Iansiti–Levien Ecosystem Patterns

SLIBRARY 03 · SHEET
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Keystone · niche · dominator · hub-landlord — four roles in any digital ecosystem.

PROFESSIONAL DEFINITION

Iansiti and Levien's taxonomy of strategic roles within a business ecosystem: Keystones provide platform value to many participants and capture a share of the surplus; Niche players specialise within a keystone's ecosystem; Dominators absorb most ecosystem value into themselves; Hub-landlords extract rent without contributing platform value.

WHO DEVELOPED IT

Iansiti, M. and Levien, R. (2004) 'The Keystone Advantage: What the New Dynamics of Business Ecosystems Mean for Strategy, Innovation, and Sustainability', Harvard Business School Press.

WHY IT WAS DEVELOPED

Conceptual lineage: Moore, J.F. (1993, 1996) 'The Death of Competition'.

FIGURE 1 · IANSITI–LEVIEN ECOSYSTEM PATTERNS



Figure 1. Iansiti–Levien Ecosystem Patterns: Keystone · niche · dominator · hub-landlord — four roles in any digital ecosystem.

HOW IT IS USED

when analysing an ecosystem's health and the firm's role within it.

WHEN IT IS USED

in M&A and partnership strategy.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Technology & Ecosystems.

REAL-WORLD EXAMPLES

- Apple App Store: keystone with elements of dominator behaviour (subject of EU DMA enforcement).
- Microsoft Azure: keystone strategy — explicit ecosystem investment.
- Twitter/X 2022+: shifted toward dominator/hub-landlord — ecosystem decline visible.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Iansiti–Levien Ecosystem Patterns contributes a clearer, more transferable framing of the dynamic it names — Keystone · niche · dominator · hub-landlord — four roles in any digital ecosystem. Its analytical contribution is the explicit recognition that keystone — creates and shares platform value (e.g. App Store giving developers reach in exchange for 15–30%).

RELATED FRAMEWORKS IN THIS COURSE

Platform vs. Pipeline · 1.3.1

Network Effects · 1.3.4

Platform Governance (Unit 3.3.)

SOURCES (HARVARD REFERENCING STYLE)

Iansiti, M. and Levien, R. (2004) The Keystone Advantage. Boston: HBR Press.

Moore, J.F. (1993) 'Predators and prey: a new ecology of competition', Harvard Business Review, 71(3), pp. 75–86.

Jacobides, M.G., Cennamo, C. and Gawer, A. (2018) 'Towards a theory of ecosystems', Strategic Management Journal, 39(8), pp. 2255–2276.

Network Effects

SLIBRARY 03 · SHEET
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Each new user makes the network more valuable — the central digital-economy mechanic.

PROFESSIONAL DEFINITION

A network effect exists when the value of a product or service to one user depends on the number of other users. Direct network effects: same-side benefit (more users → more useful, e.g. WhatsApp). Indirect (cross-side): two distinct user groups benefit each other (e.g. more Uber drivers → more passenger value). The dominant moat in digital markets.

WHO DEVELOPED IT

Term coined by Rohlfs, J. (1974) 'A theory of interdependent demand for a communications service', Bell Journal of Economics.

WHY IT WAS DEVELOPED

Popularised by Katz, M.L. and Shapiro, C. (1985, 1994). Modern platform-economics treatment in Eisenmann, Parker & Van Alstyne (2006).

FIGURE 1 · NETWORK EFFECTS

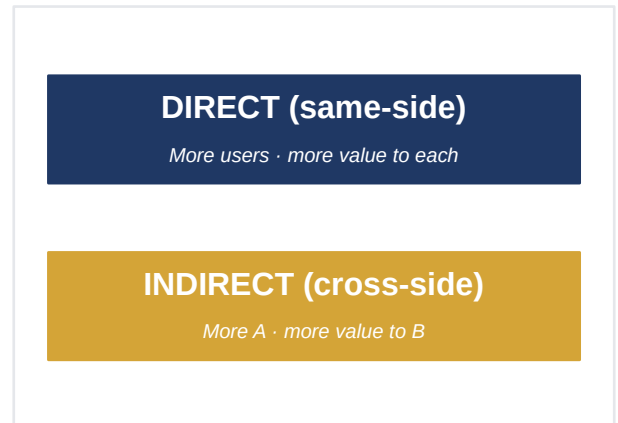


Figure 1. Network Effects: Each new user makes the network more valuable — the central digital-economy mechanic.

HOW IT IS USED

when assessing the moat strength of any platform business.

WHEN IT IS USED

in due diligence for platform M&A and venture investments.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Technology & Ecosystems.

REAL-WORLD EXAMPLES

- WhatsApp: very strong direct effect — value purely from who else is on it.
- Uber: moderate cross-side effect — strong locally, weak globally.
- Microsoft Office in 2000s: dominant via document-format network effect.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Network Effects contributes a clearer, more transferable framing of the dynamic it names — Each new user makes the network more valuable — the central digital-economy mechanic. Its analytical contribution is the explicit recognition that direct effect: $utility(user_i)$ increases with n (number of users).

RELATED FRAMEWORKS IN THIS COURSE

Metcalfe's Law (Unit 2.2.1).

Connection Density

Tipping Point · Winner-Takes-A

SOURCES (HARVARD REFERENCING STYLE)

Rohlfs, J. (1974) 'A theory of interdependent demand for a communications service', Bell Journal of Economics, 5(1), pp. 16–37.

Katz, M.L. and Shapiro, C. (1994) 'Systems competition and network effects', Journal of Economic Perspectives, 8(2), pp. 93–115.

Eisenmann, T., Parker, G. and Van Alstyne, M.W. (2006) 'Strategies for two-sided markets', Harvard Business Review, 84(10), pp. 92–101.

Super-App Model

SLIBRARY 03 · SHEET
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WeChat · Alipay · Grab — single app · many integrated services.

PROFESSIONAL DEFINITION

A single mobile application that provides many distinct services through embedded sub-apps or mini-programs — chat, payments, ride-hailing, food delivery, e-commerce, lifestyle, government services. Pioneered in Asia (WeChat 2011+, Alipay 2013+, Grab 2016+); contrast with the more fragmented Western app-store-per-service model.

WHO DEVELOPED IT

Concept articulated by BlackBerry founder Mike Lazaridis ~2010, but operationally pioneered by Tencent's WeChat with the launch of Mini Programs (2017).

WHY IT WAS DEVELOPED

Academic frame in Steinberg, M. (2019) 'The Platform Economy: How Japan Transformed the Consumer Internet'.

FIGURE 1 · SUPER-APP MODEL

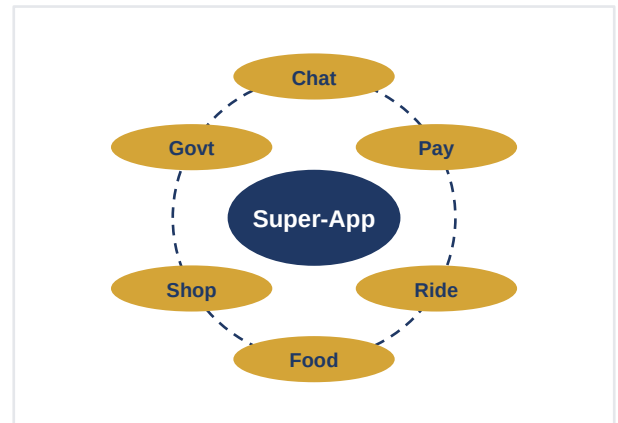


Figure 1. Super-App Model: WeChat · Alipay · Grab — single app · many integrated services.

HOW IT IS USED

when analysing competitive dynamics in mobile-commerce-heavy markets.

WHEN IT IS USED

in product strategy for emerging-market entrants.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Technology & Ecosystems.

REAL-WORLD EXAMPLES

● WeChat (Tencent): chat → payments → mini-programs → 1.3bn MAU.

● Alipay (Ant Group): payments → super-app → financial services.

● Grab (SE Asia): ride-hailing → food → financial services.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Super-App Model contributes a clearer, more transferable framing of the dynamic it names — WeChat · Alipay · Grab — single app · many integrated services. Its analytical contribution is the explicit recognition that core anchor service (chat, payments, or commerce) drives high daily-active engagement.

RELATED FRAMEWORKS IN THIS COURSE

Multi-Sided Platforms (Unit 3.)

Platform vs. Pipeline · 1.3.1

Network Effects · 1.3.4

SOURCES (HARVARD REFERENCING STYLE)

Steinberg, M. (2019) The Platform Economy: How Japan Transformed the Consumer Internet. Minneapolis: University of Minnesota Press.

Cusumano, M.A., Gawer, A. and Yoffie, D.B. (2019) The Business of Platforms. New York: HarperBusiness.

Tencent (2024) Tencent Annual Report 2024. Shenzhen: Tencent Holdings Ltd.

Christensen's Disruption

SLIBRARY 04 · SHEET
1 / 5

Lower-end entrant improves over time and overtakes incumbent leaders.

PROFESSIONAL DEFINITION

Clayton Christensen's theory of disruptive innovation: a new entrant with an inferior but cheaper product targets either a low-end segment or a non-consumer market that incumbents underserve, then improves its product over time, eventually meeting and surpassing the performance demanded by the mainstream market — at which point incumbents collapse. The pattern recurs across industries.

WHO DEVELOPED IT

Christensen, C.M. (1997) 'The Innovator's Dilemma', Harvard Business School Press.

WHY IT WAS DEVELOPED

Refined in Christensen, Raynor & McDonald (2015) 'What is Disruptive Innovation?', Harvard Business Review.

FIGURE 1 · CHRISTENSEN'S DISRUPTION

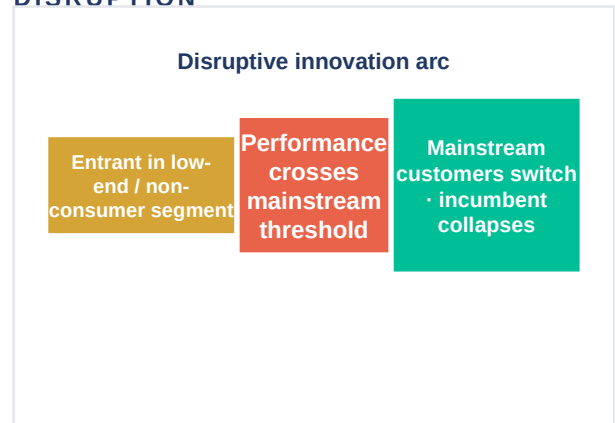


Figure 1. Christensen's Disruption: Lower-end entrant improves over time and overtakes incumbent leaders.

HOW IT IS USED

when assessing whether a new entrant is genuinely disruptive vs. merely competing.

WHEN IT IS USED

in industry-strategy work for incumbents.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Disruption Patterns.

REAL-WORLD EXAMPLES

● Personal computers vs mainframes (1980s).

● Netflix DVD-mail vs Blockbuster (1997+); Netflix streaming vs cable (2007+).

● Tesla vs traditional auto (2008+) — debated as disruption vs sustaining.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Christensen's Disruption contributes a clearer, more transferable framing of the dynamic it names — Lower-end entrant improves over time and overtakes incumbent leaders. Its analytical contribution is the explicit recognition that step 1 — Entrant launches a worse-but-cheaper or simpler product targeting an underserved or non-consumer segment.

RELATED FRAMEWORKS IN THIS COURSE

Atoms → Bits · 1.1.1

Industry Digitization Index · 1.4.2

Before/After: Music · Retail · Finance · 1.4.4

SOURCES (HARVARD REFERENCING STYLE)

Christensen, C.M. (1997) The Innovator's Dilemma. Boston: Harvard Business School Press.

Christensen, C.M., Raynor, M.E. and McDonald, R. (2015) 'What is disruptive innovation?', Harvard Business Review, 93(12), pp. 44–53.

Lepore, J. (2014) 'The disruption machine: what the gospel of innovation gets wrong', The New Yorker, June 23.

Industry Digitization Index

SLIBRARY 04 · SHEET
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MGI scoring — labour · asset · usage dimensions of industry digitisation.

PROFESSIONAL DEFINITION

McKinsey Global Institute's 2015–2020 series of indices measuring the digitisation of US and European industries along three dimensions: digital assets (digital infrastructure spending, hardware, software), digital usage (transactions, customer interactions, digital business processes), and digital labour (digital skills, digital work practices). Reveals very large cross-industry gaps.

WHO DEVELOPED IT

McKinsey Global Institute (2015) 'Digital America: A Tale of the Haves and Have-Mores'; (2018) 'Digital Europe: Pushing the Frontier, Capturing the Benefits'.

WHY IT WAS DEVELOPED

The framework was developed to give practitioners a precise, transferable vocabulary for the dynamic captured in its tagline — replacing the looser pre-existing terms used in earlier industry and academic discourse.

FIGURE 1 · INDUSTRY DIGITIZATION INDEX



Figure 1. Industry Digitization Index: MGI scoring — labour · asset · usage dimensions of industry digitisation.

HOW IT IS USED

as benchmark when assessing your industry's digital maturity.

WHEN IT IS USED

in cross-industry M&A and consulting work.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Disruption Patterns.

REAL-WORLD EXAMPLES

- Construction: persistent bottom-tier; productivity stagnant for decades.
- Healthcare: middling — heavy regulation slows digitisation.
- Banking: top-tier in payments and trading, bottom-tier in lending and back-office.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Industry Digitization Index contributes a clearer, more transferable framing of the dynamic it names — MGI scoring — labour · asset · usage dimensions of industry digitisation. Its analytical contribution is the explicit recognition that three dimensions, each scored 0–100 across ~20 indicators.

RELATED FRAMEWORKS IN THIS COURSE

OECD 3-Layer Model · 1.1.2

Christensen's Disruption · 1.4.1

BCG 10-20-70 Rule · 1.4.3

SOURCES (HARVARD REFERENCING STYLE)

McKinsey Global Institute (2015) Digital America: A Tale of the Haves and Have-Mores. New York: McKinsey & Company.

McKinsey Global Institute (2018) Digital Europe: Pushing the Frontier, Capturing the Benefits. New York: McKinsey & Company.

Brynjolfsson, E. and McAfee, A. (2014) The Second Machine Age. New York: W.W. Norton.

BCG 10-20-70 Rule

SLIBRARY 04 · SHEET
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10% tech · 20% data · 70% organisation — where transformation value really comes from.

PROFESSIONAL DEFINITION

BCG's empirical rule of thumb derived from digital-transformation research: ~10% of value comes from the technology itself, ~20% from data and analytics, ~70% from organisational and process change. The headline lesson: technology is necessary but not sufficient; transformations that under-invest in the 70% predictably underdeliver.

WHO DEVELOPED IT

BCG (2019, 2020, 2024) digital-transformation research.

WHY IT WAS DEVELOPED

Builds on Westerman, Bonnet & McAfee (2014) 'Leading Digital' (MIT-CDB), establishing that digital-mature firms outperform peers because of organisational redesign, not technology adoption.

FIGURE 1 · BCG 10-20-70 RULE

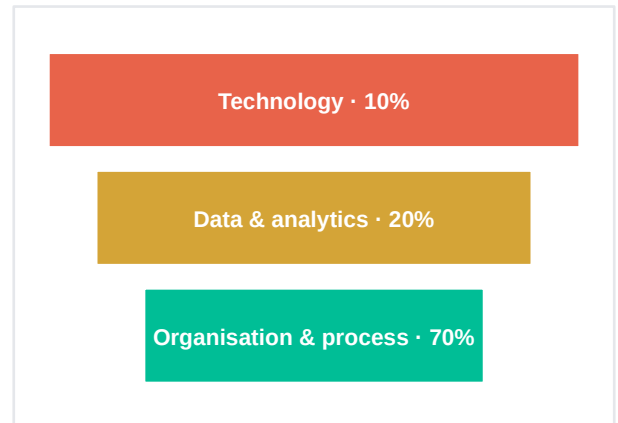


Figure 1. BCG 10-20-70 Rule: 10% tech · 20% data · 70% organisation — where transformation value really comes from.

HOW IT IS USED

when designing or evaluating a transformation business case.

WHEN IT IS USED

in board-level discussions to recalibrate where to spend.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Disruption Patterns.

REAL-WORLD EXAMPLES

- Microsoft cultural transformation (Nadella, 2014–): tech enablement followed culture change.
- DBS Bank: 'World's Best Digital Bank' invested in talent and ways-of-working before stack modernisation.
- Failed cases: many large-bank 'digital transformation' programmes burn billions on tech without addressing operating-model complexity.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, BCG 10-20-70 Rule contributes a clearer, more transferable framing of the dynamic it names — 10% tech · 20% data · 70% organisation — where transformation value really comes from. Its analytical contribution is the explicit recognition that technology layer (~10%): platforms, cloud, AI tools — genuine but commoditised contribution.

RELATED FRAMEWORKS IN THIS COURSE

Digitisation · Digitalisation · DT · 1.1.5

Industry Digitization Index · 1.4.2

McKinsey 7S

SOURCES (HARVARD REFERENCING STYLE)

BCG (2020) Flipping the Odds of Digital Transformation Success. Boston: Boston Consulting Group.

Westerman, G., Bonnet, D. and McAfee, A. (2014) Leading Digital: Turning Technology into Business Transformation. Boston: HBR Press.

BCG (2024) The Digital Acceleration Index 2024. Boston: Boston Consulting Group.

Before/After: Music · Retail · Finance

SLIBRARY 04 · SHEET
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Three textbook disruption case studies — what changed structurally.

PROFESSIONAL DEFINITION

A comparative case-based framework illustrating disruption patterns across three different industries: music (high-IP, low physical-asset intensity), retail (low-IP, medium physical-asset), and finance (regulated, very high information intensity). Each shows different disruption mechanics, different incumbent responses, and different end-states.

WHO DEVELOPED IT

Composite of three teaching cases.

WHY IT WAS DEVELOPED

Music: Knopper, S. (2009) 'Appetite for Self-Destruction'; Brynjolfsson & McAfee on Spotify (2017). Retail: Stone, B. (2013) 'The Everything Store'; Cusumano on Amazon (2019). Finance: King, B. (2018) 'Bank 4.0'; Skinner, C. (2018) 'Digital Human'.

FIGURE 1 · BEFORE/AFTER: MUSIC · RETAIL · FINANCE

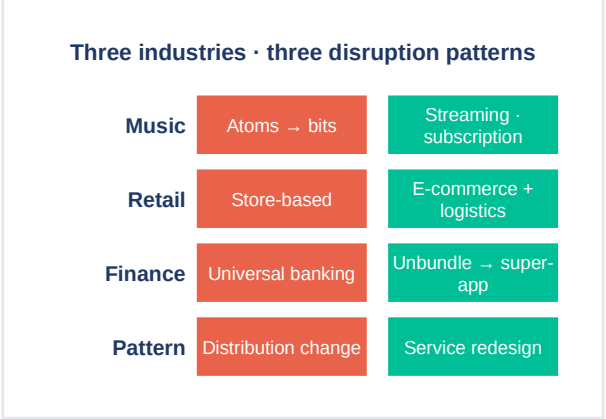


Figure 1. Before/After: Music · Retail · Finance: Three textbook disruption case studies — what changed structurally.

HOW IT IS USED

in teaching settings to illustrate that disruption patterns vary by industry structure.

WHEN IT IS USED

when asking 'how will our industry be disrupted?'

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Disruption Patterns.

REAL-WORLD EXAMPLES

- Music: Spotify, Apple Music, YouTube Music ~85% of recorded-music revenue (IFPI 2024).
- Retail: Amazon ~38% of US e-commerce; Alibaba ~50% of China e-commerce.
- Finance: Revolut, Chime, Ant — fintech share growing; incumbents partnering or acquiring.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, Before/After: Music · Retail · Finance contributes a clearer, more transferable framing of the dynamic it names — Three textbook disruption case studies — what changed structurally. Its analytical contribution is the explicit recognition that music — disruption pattern: bits replace atoms (CDs → MP3 → streaming); incumbents (labels) lose distribution control but retain rights ownership.

RELATED FRAMEWORKS IN THIS COURSE

- Christensen's Disruption · 1.4.1
- Atoms → Bits · 1.1.1
- Subscription Economy (Unit 2.3)

SOURCES (HARVARD REFERENCING STYLE)

Knopper, S. (2009) Appetite for Self-Destruction: The Spectacular Crash of the Record Industry in the Digital Age. New York: Free Press.

Stone, B. (2013) The Everything Store: Jeff Bezos and the Age of Amazon. New York: Little, Brown.

King, B. (2018) Bank 4.0: Banking Everywhere, Never at a Bank. Singapore: Marshall Cavendish.

WEF Future of Jobs

SLIBRARY 04 · SHEET
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Net new jobs created vs. displaced — the labour-market arithmetic of disruption.

PROFESSIONAL DEFINITION

The World Economic Forum's biennial Future of Jobs Report estimates how many jobs are created, destroyed, and transformed by technology adoption over a 5-year horizon, based on a survey of ~800 global employers. The 2023 edition projects ~83m jobs displaced, ~69m created — a net loss of ~14m by 2027, with substantial occupational reshuffling.

WHO DEVELOPED IT

World Economic Forum (biennial since 2016) Future of Jobs Report.

WHY IT WAS DEVELOPED

Methodology refined over time; 2023 onward incorporates explicit AI-impact categories. Conceptually grounded in Frey & Osborne (2013) 'The Future of Employment'.

FIGURE 1 · WEF FUTURE OF JOBS



Figure 1. WEF Future of Jobs: Net new jobs created vs. displaced — the labour-market arithmetic of disruption.

HOW IT IS USED

in policy and corporate workforce-planning analysis.

WHEN IT IS USED

in higher-education curriculum design.

BY WHOM IT IS USED

strategy consultants, corporate executives, policy analysts, and academic researchers working in Disruption Patterns.

REAL-WORLD EXAMPLES

- Banking sector: bank-teller decline + AI/ML and compliance specialist growth.
- Retail: cashier decline + e-commerce and logistics-coordinator growth.
- Healthcare: medical-record clerk decline + AI-assisted clinician roles growing.

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

Compared with prior approaches in the same domain, WEF Future of Jobs contributes a clearer, more transferable framing of the dynamic it names — Net new jobs created vs. displaced — the labour-market arithmetic of disruption. Its analytical contribution is the explicit recognition that survey of ~800 large global employers across ~45 countries.

RELATED FRAMEWORKS IN THIS COURSE

- Industry Digitization Index · 1.4.2
- Reskilling vs. Upskilling (Unit)
- Skills Matrix 2025-2030 (Unit)

SOURCES (HARVARD REFERENCING STYLE)

- WEF (2023) Future of Jobs Report 2023. Geneva: World Economic Forum.
- WEF (2025) Future of Jobs Report 2025. Geneva: World Economic Forum.
- Frey, C.B. and Osborne, M.A. (2013) The Future of Employment: How Susceptible Are Jobs to Computerisation? Oxford: Oxford Martin School.